

Theory Week 8

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Orga

Mechanics 3 CAB G59
WhatsApp group



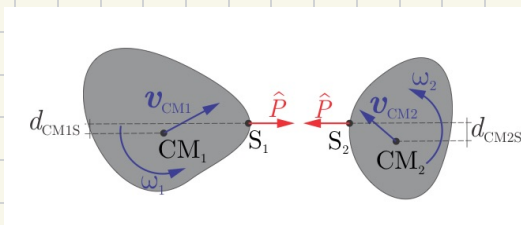
- Find all doc's on my website (link in WhatsApp bio)
- remember to do the STACK
- Study Center
→ Wednesday 18:00-20:00 (ETC &)

8	04/11/23*	rigid body collisions	3.3
	06/11/23*	active vs. passive rotations, LMB in moving frames	3.4-3.4.2
		exercise 8: rigid-body collisions, rotations	
midterm		(voluntary) problem set via Moodle/STACK in week 8	

3.3 rigid body collisions

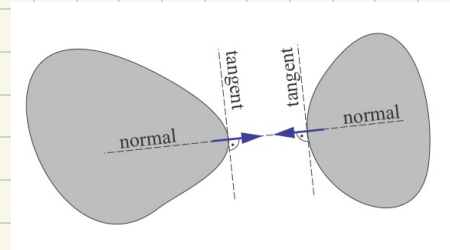
eccentric collision:

collision between bodies, where impulsive forces are exchanged, but not on the line connecting the CMs of the two bodies.



oblique collision:

if impulsive forces exchange is at an angle to the line of impact.



Derivation in Lecture $\Rightarrow \hat{P} = \text{impulse force} = \int_{t_-}^{t_+} P(t) dt$



collision of rigid bodies: linear and angular momentum balance with contact point S:

$$\begin{aligned} \hat{P} &= m_1 [v_{CM1}(t_+) - v_{CM1}(t_-)], & r_{CM1S} \times \hat{P} &= I_{CM1} [\omega_1(t_+) - \omega_1(t_-)] \\ -\hat{P} &= m_2 [v_{CM2}(t_+) - v_{CM2}(t_-)], & r_{CM2S} \times (-\hat{P}) &= I_{CM2} [\omega_2(t_+) - \omega_2(t_-)] \end{aligned}$$

Differentiate ^(frictionless) smooth & rough bodies:

①

collision of two smooth, **frictionless** bodies:

$$v_{CM1}^t(t_+) = v_{CM1}^t(t_-), \quad v_{CM2}^t(t_+) = v_{CM2}^t(t_-)$$

→ momentum transferred only in normal dir.

②

collision of two rough bodies, assuming **no slip**:

$$v_{S1}^t(t_+) = v_{S2}^t(t_+)$$

→ momentum exchanged in both normal & tangential dir.

As always: we also need the coeff. of rest. e

coefficient of restitution for a two-body collision (contact points S_1 and S_2):

$$e = -\frac{v_{S2}^n(t_+) - v_{S1}^n(t_+)}{v_{S2}^n(t_-) - v_{S1}^n(t_-)}$$

for a general rigid body collision problem you always need to combine these formulas + use velocity transfer formula

3.4 non-inertial frames

- Def:
- Inertial frame : An absolute frame of reference, whose basis & origin are time-indep.
 - Inertial ref. frame : A frame that is fixed / not moving
 - Non-inertial frame : A non-inertial frame is a moving frame of reference whose basis &/or origin depend on time.

T^{CM} = rotation matrix that converts vector components from M -basis to to corresponding components in the C -basis

orthogonal
passive rot.
↳ vectors don't move (only change basis)

$R^{CM} = R^{M \leftarrow C}$: Takes the C -basis vectors $\{\underline{e}_1^C, \underline{e}_2^C, \underline{e}_3^C\}$ & rotates them onto M -basis vectors $\{\underline{e}_1^M, \underline{e}_2^M, \underline{e}_3^M\}$

orthogonal $\rightarrow R^{CM} = (R^{MC})^T$ since $R^T = R^{-1}$ (remember Unitary :))

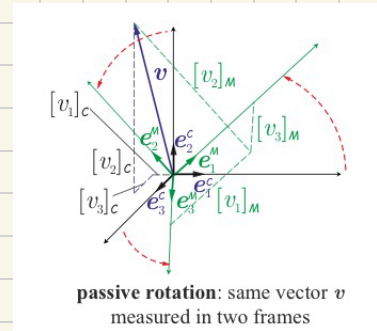
active rot.
↳ takes $\underline{v}$ & rotates onto $\underline{u} = R\underline{v}$

$$\begin{cases} \underline{e}_1^M = R^{MC} \underline{e}_1^C \\ \underline{e}_2^M = R^{MC} \underline{e}_2^C \end{cases} \quad \underline{e}_3^M = R^{MC} \underline{e}_3^C$$

Passive rotations

passive rotation between the M - and C -frames:

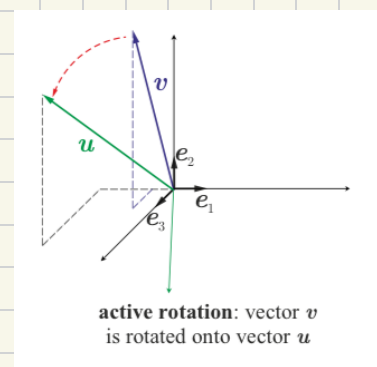
$$[v]_C = \sum_{j=1}^3 [R_{ij}^{CM}]^T [v]_M, \quad [v]_M = \sum_{j=1}^3 [R_{ij}^{MC}]^T [v]_C \quad \text{with} \quad [R_{ij}^{MC}] = [R_{ji}^{CM}] = \underline{e}_i^C \cdot \underline{e}_j^M$$



Active rotations

active rotations:

$$\underline{e}_i^M = R^{MC} \underline{e}_i^C, \quad \underline{e}_i^C = R^{CM} \underline{e}_i^M \quad \text{with} \quad R^{MC} = (R^{CM})^T$$



time derivative relation between an inertial frame C and a moving frame M :

try to derive polar coordinate acc. using this formula!

rotating: $\dot{\underline{r}}^C = \dot{\underline{r}}^M + \underline{\Omega}^M \times \underline{r}$

general: $\dot{\underline{r}}_P^C = \dot{\underline{r}}_{OP}^M + \dot{\underline{r}}_{OM}^M + \underline{\Omega}^M \times \underline{r}_{OM}^M$

vel. seen from outside obs. (not in moving frame)
vel. seen from moving frame
"correlation terms"

elementary rotations in 3D about the $\underline{e}_i$ -axis by angle φ_i :

$$[R^{M_1C}]_C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi_1 & -\sin \varphi_1 \\ 0 & \sin \varphi_1 & \cos \varphi_1 \end{pmatrix},$$

$$[R^{M_2C}]_C = \begin{pmatrix} \cos \varphi_2 & 0 & \sin \varphi_2 \\ 0 & 1 & 0 \\ -\sin \varphi_2 & 0 & \cos \varphi_2 \end{pmatrix},$$

$$[R^{M_3C}]_C = \begin{pmatrix} \cos \varphi_3 & -\sin \varphi_3 & 0 \\ \sin \varphi_3 & \cos \varphi_3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$